

- 5 The photograph shows a humphead cichlid fish found only in Lake Tanganyika in Africa.



(Source: © Martin Lindsay/Alamy Stock Photo)

A behavioural adaptation shown by the female humphead cichlid is that fertilised egg cells are kept in her mouth until they hatch.

- (a) (i) Which of the following is a term used to describe the fact that this fish is found only in Lake Tanganyika?

(1)

- ☐ A diversity
- ☐ B endemic
- ☐ C homozygous
- ☐ D specific

- (ii) Suggest why the female humphead cichlids have this behavioural adaptation.
Give a reason for your answer.

(2)

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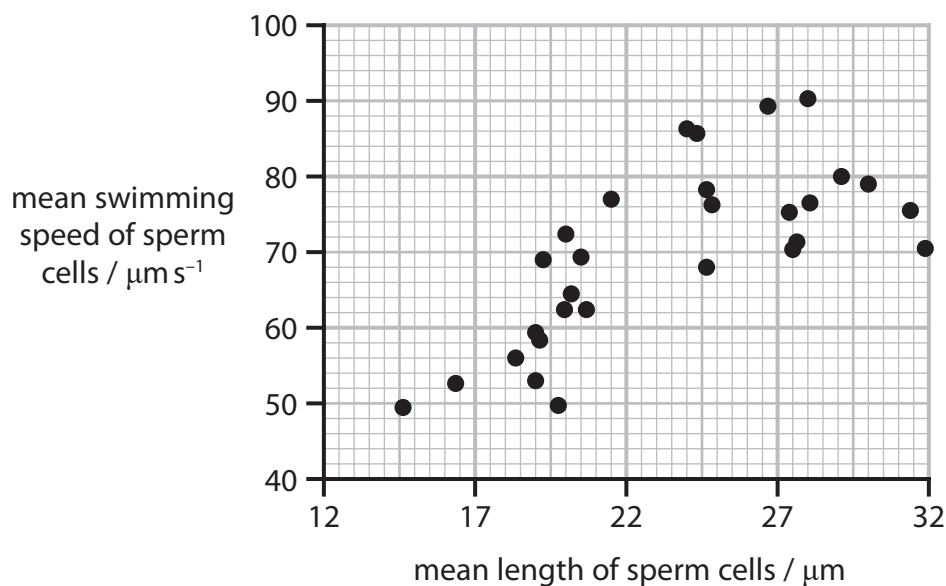
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- (b) The relationship between the length and swimming speed of the sperm cells of 30 cichlid species was investigated.

The graph shows the results of this investigation.



- (i) State the relationship between the mean length and swimming speed of the sperm cells of these cichlid species.

(1)

- (ii) A scientist wanted to assess the strength of the relationship between these two variables.

Which statistical test should be used to assess the strength of the relationship between these two variables?

(1)

- ☐ A chi-squared test
- ☐ B correlation coefficient
- ☐ C index of diversity
- ☐ D Student's t-test

- (iii) Suggest **one** advantage for a male cichlid of producing sperm cells with a faster swimming speed.

(1)

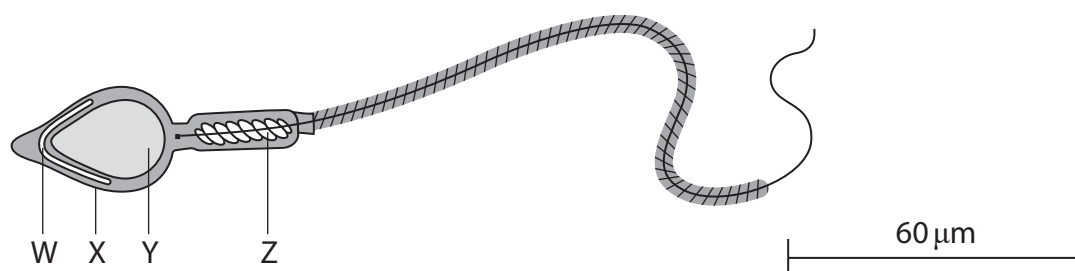
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- (c) A sperm cell contains an acrosome, a specialised lysosome.

The diagram shows a sperm cell.



- (i) Which labelled structure is an acrosome?

(1)

- ☐ **A** W
- ☐ **B** X
- ☐ **C** Y
- ☐ **D** Z

- (ii) Calculate the magnification of the sperm cell shown in the diagram.

Give your answer to **two** significant figures.

(2)

Answer



- (iii) Compare and contrast the structure and function of lysosomes and acrosomes.

(3)

(Total for Question 5 = 12 marks)

